



ASL-BO: Bayesian Optimization with Large Language Models for Analog Circuit Sizing

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Abstract. With the continuous development of analog integrated circuits, machine learning is increasingly being applied in circuit design to reduce the growing difficulty and complexity of design. *Bayesian Optimization* (BO) is a representative method applied to analog circuit sizing, treating sizing as a black-box numerical optimization problem. However, existing BO faces challenges of limited stability and slow convergence. To address these issues, we constructed an LLM-based guider agent within Bayesian Optimization, and proposed ASL-BO, an LLM agent enhanced BO method for circuit sizing. The guider agent analyzes topological equivalence to reduce parameters, generates high-quality initial samples, dynamically adjusts sampling bounds via in-context learning, and adaptively selects optimal sampling schemes from an acquisition functions pool. Validated on operational amplifier and hysteresis comparator across four specifications, ASL-BO outperforms baselines significantly. It largely improves sizing success rate and reduces simulation cost, outperforming baselines. Ablation study and discussion on the evolution of FoM (Figure of Merit) confirm the effectiveness of each component and highlight ASL-BO's robustness and efficiency in analog circuit sizing.

Keywords: Analog Circuit Sizing · Large Language Models · Bayesian Optimization · Electronic Design Automation · Machine Learning

1 Introduction

Analog circuits are circuits that process continuously-valued signals in a pre-designed manner [12, 14, 25]. They perform operations such as addition, sampling, filtering, and binarization on analog signals to support subsequent signal processing. This precise signal conversion constitutes the design foundation of modern analog integrated circuits, which is particularly crucial for ensuring circuit performance as integrated circuit manufacturing processes move into the deep submicron era [1].

With the continuous expansion of the scale of analog integrated circuits, the difficulty and complexity of circuit design are increasing. Relying solely on manual design

by human engineers has become inefficient. The development of machine learning has greatly improved this situation [10,34,35]. As in the circuit design flow shown in Fig. 1, sizing can be modeled as a numerical optimization problem of *figure of merit* (FoM) and machine learning can be applied to it, which is the key to ensuring circuit validity and requires substantial manual design experience and time [12]. Methods such as differential evolution [27] and covariance matrix adaptation evolution strategy [26] have been successfully applied in sizing, among which Bayesian optimization (BO) is the most representative [21,22]. BO derives the posterior distribution from the given prior distribution and observed data, and randomly samples from the posterior distribution to approximate its characteristics, which makes it of great potential in complex circuit sizing.

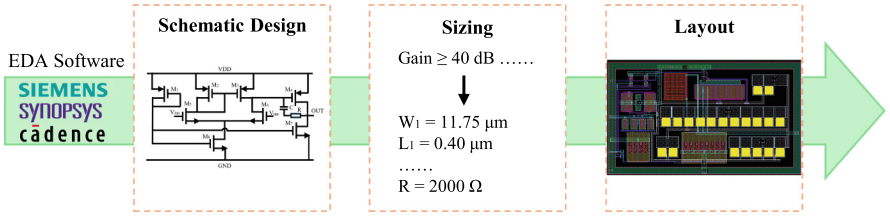


Fig. 1. Overview of analog circuit design flow. In sizing, circuit parameters are determined under a fixed topology to make the circuit performance meets specifications.

However, the current BO in analog circuit sizing still has challenges in two aspects:

- (1) **Limited stability:** As a black-box optimization algorithm, BO only captures partial correlations between parameters and FoM. When prior parameters fail to meet basic requirements, the circuit will malfunction, resulting in a lack of valid information in FoM, which makes its performance highly related to initial random sampling.
- (2) **Slow convergence:** Given the intricate relationship between analog circuit parameters and performance metrics, BO fails to accurately model the posterior distribution with limited sampling iterations due to the lack of prior knowledge about analog circuit. It exhibits slow convergence and consumes excessive simulation computing resources. These are also the challenges faced by other similar machine algorithms applied in analog circuit sizing.

Recently, neural networks [18,30,31] and large language models (LLMs) have achieved remarkable advancements, boasting profound knowledge of analog circuits and good capacity for circuit analysis [28]. Introducing LLMs into BO through specific mechanisms is expected to solve these challenges and yield improvements. However, previous studies have indicated that LLMs are not good at precise numerical computation, while BO excels at handling numerical problems [32]. Therefore, a collaborative method is required to fully harness the complementary strengths of LLMs and BO and apply them to analog circuit sizing. In previous work on ADO-LLM [33], the researchers have recognized the value of integration, yet merely treating LLMs as a sampler clearly fails to maximize their potential.

In this way, we construct an LLM-based guider agent, integrate it with BO, and propose **ASL-BO** (Analog Circuit Sizing LLM Agent enhanced Bayesian Optimization Method). Through prompt engineering, the agent analyzes the topological equivalence to reduce the parameters, generate initial samples and adjust the sampling space according to simulation results. Multiple Gaussian Process-based Bayesian Optimization proposers (GP-BO) with distinct strategy functions propose sampling schemes within the range. The agent then selects one sampling scheme for simulation based on historical information. Through in-context learning, simulation results are further sent to the agent to adjust sampling bounds, and the above process is repeated. Through the effective integration of the guider agent and BO, ASL-BO improves the sizing efficiency of BO in analog circuit sizing.

To validate the efficiency of ASL-BO, we choose operational amplifier (op-amp) and hysteresis comparator for sizing, which have 18 and 16 independent parameters respectively and both were sized for four different specifications. In this paper, we make two main contributions:

1. We propose ASL-BO, an LLM agent enhanced Bayesian Optimization method for analog circuit sizing, which significantly improves the analog circuit sizing efficiency of Bayesian Optimization.
2. We conduct validation on two typical analog circuits with four specifications. Results show that ASL-BO achieves improvements over baselines by more than **65%** in FoM, success rates and stimulation cost.

2 Related Work

2.1 ML-Based Method in Analog Circuit Design

With the rapid development of artificial intelligence, machine learning (ML) have been employed to build fast surrogate models that map circuit performance to design specifications, thereby significantly reducing the computational cost of repeated simulations [17, 24]. These methods have demonstrated potential in tackling multi-objective optimization and constraint-satisfaction problems, thereby enhancing the automation level of the design flow [6, 16]. Most ML models heavily depend on large volumes of high-quality data, and their generalization capability remains limited when dealing with high-dimensional, strongly nonlinear circuit problems.

BO has emerged as the preferred method for analog sizing due to its ability to handle expensive simulation budgets [4, 22]. BO constructs a probabilistic *surrogate model*, typically a Gaussian Process (GP) [23], to approximate the circuit's performance landscape. The workflow of BO is summarized in Algorithm 1. However, standard BO is inefficient when applied to complex analog circuits with dozens of variables [7]. Although techniques like high-dimensional BO (tSS-BO) [13] have been proposed to mitigate this, conventional frameworks typically employ a static acquisition strategy throughout the optimization campaign. Such black-box optimization fails to adapt to the growing complexity and difficulty of analog circuit sizing.

Algorithm 1. Bayesian Optimization for Analog Circuit Sizing**Require:** Initial Data $\mathcal{D}$, Acquisition Function α

- 1: **while** $t < T_{max}$ and Best($\mathcal{D}$) not meets $\mathcal{S}$ **do**
- 2: Fit probabilistic model $\mathcal{G} \mathcal{P}$ to observed data $\mathcal{D}$
- 3: $\mathbf{x}_{next} \leftarrow \operatorname{argmax}_{\mathbf{x} \in \Omega} \alpha(\mathbf{x} | \mathcal{G} \mathcal{P})$ ▷ Maximize Acquisition
- 4: $y_{next} \leftarrow \text{FoM}(\text{Simulate}(\mathbf{x}_{next}))$ ▷ Evaluate Samples
- 5: $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{x}_{next}, y_{next})\}$
- 6: **end while**

2.2 Large Language Models in Analog Circuit Design

The advent of LLMs has introduced semantic reasoning capabilities to the EDA workflow [9, 19]. Beyond general code generation [20], recent works such as Analog-Coder [15] demonstrate that LLMs can generate SPICE netlists and simulation scripts from natural language prompts. To bridge the gap between semantic understanding and numerical precision, ADO-LLM [33] proposed a hybrid architecture that integrates LLMs with Bayesian Optimization. ADO-LLM leverages the LLM to generate initial designs and one sample in each sampling batch. It adheres to a static acquisition function and treats the LLM primarily as a smart sampler rather than a decision-maker or tool-user. The synergy between the semantic high-level planning of LLMs and the numerical search of BO remains underutilized in analog circuit sizing.

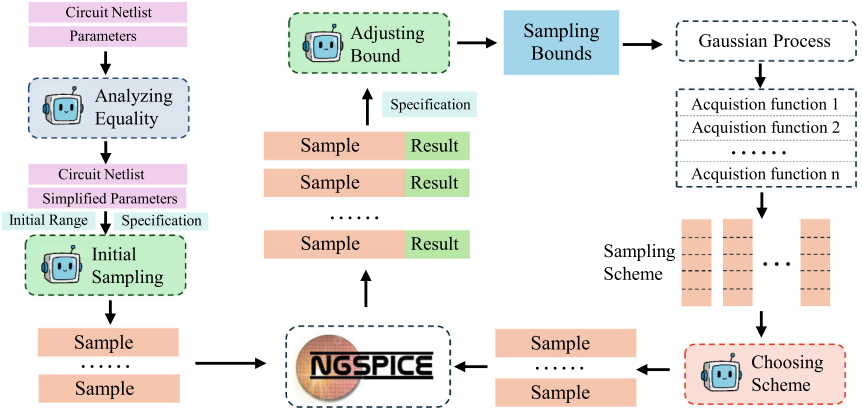
3 Method

Fig. 2. Overview of ASL-BO method. The guider agent in blue box indicates that it does not participate in in-context learning; the agents in green box indicates that they uses the context and incorporates their dialogues into the context; the agent in red box only uses the context without modifying it. (Color figure online)

As shown in Fig. 2, ASL-BO combines LLM-based guider agent with BO framework. This method consists of four primary stages: (1) **Topological equivalence analysis and initialization**; (2) **Dynamic parameter bounds adjustment**; (3) **Adaptive sampling scheme choosing**; and (4) **Batch simulation with closed-loop iterative optimization**. The workflow of ASL-BO is summarized in Algorithm 2. Beyond merely executing optimization tasks, the agent is capable of perceiving the real-time optimization state. It dynamically adjusts the parameter bounds and the Bayesian acquisition strategy in a closed-loop manner.

3.1 Topological Equivalence Analysis and Initialization

Traditional optimization methods typically perform an agnostic search within the raw high-dimensional parameter space, often leading to inefficiencies. Our approach first leverages the agent's circuit comprehension capabilities to perform dimensionality reduction on the design space. Furthermore, it utilizes the LLM's pre-trained heuristic knowledge [8] to generate high-quality initial design points. Initially, the LLM agent acquires the SPICE netlist as input. By analyzing the circuit topology, the agent identifies critical sub-blocks such as differential pairs and current mirrors. It extracts structural equality constraints (e.g., $W_{M1} = W_{M2}$, $L_{M1} = L_{M2}$) to establish a mapping $f: \mathbb{R}^D \rightarrow \mathbb{R}^d$ ($d < D$), subsequently rewriting the SPICE netlist. This process significantly reduces the search dimensionality for Bayesian Optimization. Then it leverages the design specifications and process constraints to propose N_{init} sets of initial parameter candidates, effectively bypassing unproductive random exploration.

3.2 Dynamic Parameter Space Refinement

Previous studies typically rely on static parameter bounds, which restrict optimization flexibility and often result in excessively large search spaces [22, 33]. In ASL-BO, the guider agent dynamically adjusts the parameter bounds based on historical sampling results with In-Context Learning (ICL) [5]. At each iteration t , the agent analyzes the distribution trends of the current dataset $\mathcal{D}_t = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$. If low-performance samples are densely clustered along specific parameter dimensions, the agent suggests contracting the search boundaries to focus on high FoM regions. If the optimal solution lies near the current boundary, the agent leverages circuit principles to infer potential superior regions and dynamically extends the parameter range. Consequently, the search space Ω_t evolves throughout the iterations, significantly accelerating the fitting efficiency of the GP model.

3.3 Adaptive Sampling Scheme Choosing

Strategy Pool. Traditional BO typically employs a single acquisition function throughout the entire process, overlooking the distinct advantages of various functions in different mathematical contexts. In ASL-BO, we define the acquisition function pool as

$$C_{pool} = \{\alpha_{qLogEI}, \alpha_{qPI}, \alpha_{qKG}, \alpha_{qLogNoisyEI}, \alpha_{qUCB}\},$$

Algorithm 2. Analog Circuit Sizing LLM-enhanced Bayesian Optimization Method**Require:** Circuit Netlist $\mathcal{N}$, Design Specs $\mathcal{S}$, Parameter Bounds Ω_{init} , LLM Agent $\mathcal{M}$ **Ensure:** Optimal Circuit Parameters $\mathbf{x}^*$

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1: Step 1: Topology Analysis
2: Equivalence Map  $\mathcal{E} \leftarrow \mathcal{M}.\text{Analyze}(\mathcal{N})$  ▷ Prompt 1
3:  $\mathcal{N}', \Omega \leftarrow \text{ApplyMap}(\mathcal{N}, \Omega_{init}, \mathcal{E})$ 
4: Step 2: Initialization
5:  $\mathcal{D} \leftarrow \emptyset$ 
6:  $\mathbf{X}_{init} \leftarrow \mathcal{M}.\text{SuggestInit}(\mathcal{N}', \mathcal{S}, \Omega)$  ▷ Prompt 2
7: for  $\mathbf{x}_i$  in  $\mathbf{X}_{init}$  do
8:    $\mathbf{p}_i \leftarrow \text{Simulate}(\mathcal{N}', \mathbf{x}_i)$  ▷ Get Metrics
9:    $y_i \leftarrow \text{FOM}(\mathbf{p}_i)$ 
10:   $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{x}_i, y_i)\}$ 
11: end for
12: Step 3: Optimization Loop
13: for  $t = 1$  to  $T_{\max}$  do
14:   $\Omega_t \leftarrow \mathcal{M}.\text{UpdateBounds}(\mathcal{D}, \Omega_{t-1})$  ▷ Prompt 3
15:  Fit Gaussian Process model  $\mathcal{G} \mathcal{P}$  on  $\mathcal{D}$ 
16:   $\alpha_k \in \{qLogEI, qPI, qKG, qLogNoisyEI, qUCB\}$ 
17:   $C_{pool} \leftarrow \{\text{OptimizeAcq}(\mathcal{G} \mathcal{P}, \alpha_k, \Omega_t) \mid k = 1 \dots 5\}$ 
18:   $k^* \leftarrow \text{SelectStrategy}(C_{pool}, \mathcal{D})$ 
19:   $\mathbf{X}_{batch} \leftarrow \mathcal{M}.k^*$  ▷ Prompt 4
20:  for  $\mathbf{x}_j$  in  $\mathbf{X}_{batch}$  do
21:     $\mathbf{p}_j \leftarrow \text{Simulate}(\mathcal{N}', \mathbf{x}_j)$ 
22:     $y_j \leftarrow \text{FOM}(\mathbf{p}_j)$ 
23:  end for
24:   $\mathcal{D} \leftarrow \mathcal{D} \cup \{(\mathbf{X}_{batch}, \mathbf{Y}_{batch})\}$ 
25:  if  $\max(\mathbf{Y}_{batch})$  meets  $\mathcal{S}$  then
26:    break
27:  end if
28: end for
29: return Best  $\mathbf{x}^*$  from  $\mathcal{D}$ 

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where these functions are designed for different batch optimization scenarios, as detailed in Table 1. Each function acquires a batch of samples as a sampling scheme independently.

Table 1. Complementary Acquisition Function Strategies in the Strategy Pool

Strategy	Core Feature	Applicable Scenario
α_{qLogEI} [2]	Numerically stable gradients via Log-Sum-Exp	General-purpose noiseless optimization
α_{qPI} [11]	Maximizes likelihood of exceeding best value	Feasibility constraints & binary outcomes
α_{qKG} [3]	Maximizes value of information	Initial exploration phase
$\alpha_{qLogNoisyEI}$ [2]	Strong robustness to noise	Optimization with simulation noise
α_{qUCB} [29]	Explicit exploration control via β	User-defined risk preference tasks

Choosing Mechanism. In ASL-BO, LLM agent dynamically selects the most appropriate sampling scheme from the schemes in pool. Agent gets the historical circuit performance and the spatial distribution of existing samples through ICL with schemes and respective acquisition function in prompt. Agent is required to provide not only the chosen scheme but also the underlying reason, thereby ensuring the explainability and quality of the decision-making process.

4 Experiments

4.1 Settings

To minimize random errors and improve the reliability of the results, we performed five repetitions for each experimental group.

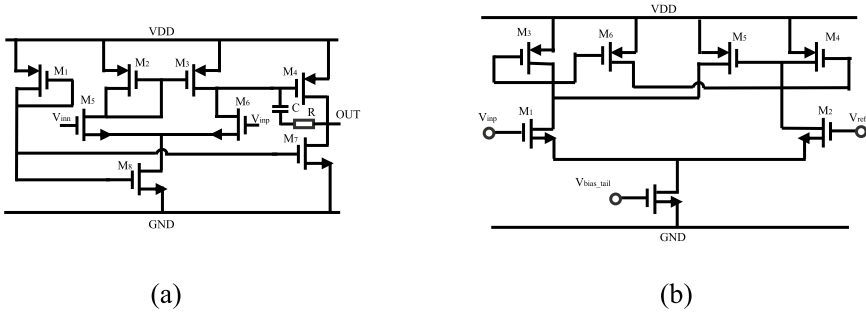


Fig. 3. Schematic of analog circuits in experiment.

Analog Circuits. We selected the Two-Stage Miller Compensation Operational Amplifier (op-amp) and the Hysteresis Comparator for sizing. Both are sized based on the SMIC 180nm CMOS RF process (SMIC 18mmRF).

As shown in Fig. 3 (a), Two-Stage Miller Compensation Operational Amplifier is a differential-input linear amplifier characterized by high gain, high input impedance, and low output impedance, and is widely used in analog circuits for signal conditioning, filter circuits, voltage comparators, oscillators, and precision measurement systems. It has 18 variable parameters and parameter ranges are set as follows: transistor width $W \in [1\mu m, 100\mu m]$, length $L \in [180nm, 10\mu m]$, compensation resistance $R \in [100\Omega, 10k\Omega]$, and compensation capacitance $C \in [0.1pF, 100pF]$.

As shown in Fig. 3 (b), Hysteresis Comparator is voltage comparator incorporating positive feedback, which sets upper and lower threshold voltages to effectively suppress frequent output switching caused by minor fluctuations in the input signal. It has 16 variable parameters and parameter ranges are set as follows $W \in [2\mu m, 100\mu m]$ and length $L \in [180nm, 2\mu m]$.

Specifications. To quantify algorithm performance, we set specifications at different difficulty levels for both circuits:

Operational Amplifier: **S-1** (Basic Specification): $Gain \geq 40dB$, $GBW \geq 10MHz$, $PM \geq 60^\circ$, $Power \leq 1mW$. **S-2** (Advanced Specification): $Gain \geq 50dB$, $GBW \geq 20MHz$, $PM \geq 90^\circ$, $Power \leq 1mW$.

Hysteresis Comparator: **S-1** (Basic Specification): $V_{hys} = 100mV \pm 5\%$, $Power \leq 0.25mW$, $Delay \leq 5.0ns$, $V_{offset} \leq 10.0mV$. **S-2** (Advanced Specification): Target Hysteresis Window $V_{hys} = 100mV \pm 3\%$, $Power \leq 0.23mW$, $Delay \leq 3.0ns$, $V_{offset} \leq 10.0mV$.

Metrics. We constructed a comprehensive FoM to guide Bayesian Optimization. The FoM is calculated as:

$$FoM_{opamp} = \widehat{Gain} + \widehat{GBW} + \widehat{PM} + (1 - \widehat{Power}) \quad (1)$$

$$FoM_{comp} = 3 + |\widehat{V_{hys}}| + (1 - \widehat{Power}) + (1 - \widehat{Delay}) + (1 - \widehat{offset}) \quad (2)$$

The $\widehat{m}$ can be calculated as:

$$\widehat{m} := \begin{cases} 1 & \text{if } m \text{ meets } m_{spec} \\ \frac{m}{m_{spec}} & \text{otherwise} \end{cases} \quad (3)$$

where m_i represents the i -th performance metric, m_{spec} represents the specification. If a circuit metric does not exist in the simulation, then set the FoM to the default value ($FoM_{opamp} = -10$, $FoM_{comp} = -15$).

Other metrics are as follows: #Missed Spec.: The number of design specifications that were not met, used to measure the gap between the final performance and specifications. #Success: The number of independent success sizing in 5 repetitions that successfully satisfied all design specifications. #Sim. Cost: The number of simulation iterations required to meet the specifications. This metric only exists upon success.

Baselines. ASL-BO has great scalability and can be constructed on various open-source or closed-source LLMs. In experiment, we compared it against the following baselines on DeepSeek-V3.2 and GPT-4.1 as they are representative commercial LLMs: GP-BO [22]: Standard Bayesian Optimization. Built upon the BoTorch and GPyTorch libraries, it employs a Single Task Gaussian Process as the surrogate model with default acquisition function α_{qLogEI} . ADO-LLM [33]: An existing LLM-assisted optimization method, used as a comparison to verify the stability of our dynamic guidance strategy.

4.2 Main Experiments

The results of main experiments, shown in Table 2 and Table 3, demonstrate the superior performance of ASL-BO across different circuit topologies and difficulty levels. DS and GPT means using Deepseek-V3.2 and GPT-4.1. The units of the metrics in table are as follows: Gain (dB), GBW (MHz), PM ($^\circ$), Power (mW), V_{sys} and V_{offset} (mV), Delay (ns). For the op-amp, ASL-BO is the only method to achieve a 100% success

rate under S-1 specifications and breaks the zero-success bottleneck of baselines in the challenging S-2 task. For the Hysteresis Comparator, ASL-BO significantly enhances optimization efficiency. Compared to GP-BO and ADO-LLM, ASL-BO reduces simulation costs by approximately 40%–65% while maintaining high FoM scores. This indicates that ASL-BO can rapidly lock onto sensitive feasible regions with minimal computational resources through dynamic boundary adjustment and adaptive strategies. Both DeepSeek and GPT-4 variants exhibit consistent high performance, validating the robustness of ASL-BO. Furthermore, the LLM computational cost is almost negligible compared to the great reduction in simulation, especially in larger circuits in the future.

Table 2. Results of main experiments. Performance of ASL-BO on op-amp sizing tasks.

Spec.	Method	Gain	GBW	PM	Power	FoM	#Missed Spec.	#Success	#Sim. Cost
S-1	GP-BO	15.9	6.85	81.9	0.47	1.45	1.8	0	—
	ADO-LLM(DS)	29.2	6.98	80.3	0.57	2.38	1.6	0	—
	ADO-LLM(GPT)	26.90	26.04	92.03	0.67	2.64	1.2	1	40
	ASL-BO(DS)	42.8	17.6	68.1	0.73	3.00	0	5	20
	ASL-BO(GPT)	46.34	27.54	127.0	0.80	3.00	0	5	24
S-2	GP-BO	21.2	6.44	106.9	0.63	1.52	2.2	0	—
	ADO-LLM(DS)	34.53	9.65	82.25	0.65	2.07	2.6	0	—
	ADO-LLM(GPT)	35.77	70.15	62.98	0.70	2.13	2	0	—
	ASL-BO(DS)	52.82	24.3	84.7	0.89	2.76	1.4	1	30
	ASL-BO(GPT)	43.15	41.19	111.4	0.59	2.79	1.4	1	35

Table 3. Results of main experiments. Performance of ASL-BO on hysteresis comparator sizing tasks.

Spec.	Method	V_{sys}	V_{offset}	Delay	Power	FoM	#Missed Spec.	#Success	#Sim. Cost
S-1	GP-BO	142.53	14.12	2.80	0.16	0.15	2	1	58
	ADO-LLM(DS)	110.94	8.60	4.94	0.24	0.23	2	2	36
	ADO-LLM(GPT)	109.80	8.72	5.04	0.24	0.21	2	1	38
	ASL-BO(DS)	97.94	5.14	2.58	0.19	1.00	0	5	19
	ASL-BO(GPT)	99.07	4.32	2.61	0.16	0.99	0.2	4	22
S-2	GP-BO	168.42	18.75	2.77	0.15	-0.12	2	0	—
	ADO-LLM(DS)	103.16	8.70	2.94	0.23	-0.01	2.6	1	58
	ADO-LLM(GPT)	103.62	8.76	3.10	0.23	-0.02	2.4	1	55
	ASL-BO(DS)	95.29	5.14	2.51	0.19	0.70	0.8	2	29
	ASL-BO(GPT)	97.81	4.86	2.57	0.17	0.99	0.8	3	30

4.3 Ablation Study

Results of the ablation study shown in Table 4 demonstrate the contributions of each component in ASL-BO. We use Deepseek-V3.2 for experiments in ablation. Topological Equivalence Analysis (w/o Equality) proves fundamental; removing it results in total failure for both circuits, confirming the necessity of dimensionality reduction. Dynamic Boundary Adjustment (w/o Bounds) shows the most significant impact, particularly for the Hysteresis Comparator which is sensitive to parameter ratios. Removing this module causes the simulation cost to surge, with a sharp drop in success rate, confirming that dynamically narrowing the search space is critical for solving high-dimensional sensitive parameters. Finally, Adaptive Strategy Choosing (w/o Choosing) effectively accelerates convergence; its absence leads to an increase in average simulation costs by approximately 50%–80%.

Table 4. Results of ablation study.

Spec.	Method	FoM	#Missed Spec.	#Success	#Sim. Cost
S-1(op-amp)	w/o Equality	2.39	1.4	0	—
	w/o Init.	2.51	1.4	0	—
	w/o Bounds	2.94	0.6	3	33
	w/o Choosing	2.99	0.2	4	30
	ASL-BO	3.00	0	5	20
S-1(camp)	w/o Equality	0.27	1	0	—
	w/o Init.	0.90	0.4	3	42
	w/o Bounds	0.89	0.8	1	50
	w/o Choosing	0.99	0.2	4	35
	ASL-BO	1.00	0	5	19

4.4 Evolution of FoM

To intuitively analyze the optimization dynamics and convergence behaviors, we analyze the FoM evolution curves of the three methods on both circuits in Fig. 4. Figure 4(a)-(c) corresponds to the Op-Amp task, while Fig. 4(d)-(f) represents the Hysteresis Comparator task. The columns from left to right correspond to ASL-BO, ADO-LLM, and GP-BO respectively.

As shown in Fig. 4(a), ASL-BO exhibits an immediate and steep ascent in the Op-Amp, locking onto the high-performance region within the first 5 iterations. In stark contrast, GP-BO, shown in Fig. 4(c), suffers from a significant cold start period, stagnating at a low FoM value for nearly 20 iterations before finding a valid design point. ADO-LLM, shown in Fig. 4(b), achieves successful initialization, but the LLM’s knowledge is not effectively transferred to BO, leading to unstable subsequent sampling.

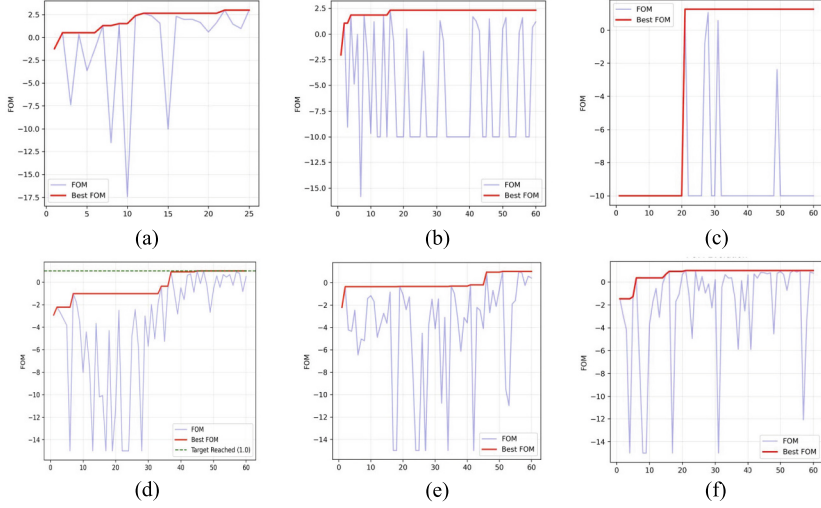


Fig. 4. Comparison of FoM evolution curves.

The advantages of ASL-BO are even more pronounced in the Hysteresis Comparator task, which involves sensitive narrow-band constraints. ASL-BO Fig. 4(d) demonstrates a robust and steady climbing trajectory, successfully reaching the target. Conversely, ADO-LLM Fig. 4(e) displays severe oscillations in FoM of samples, indicating that its static search boundaries fail to effectively constrain the exploration within the valid hysteresis region. GP-BO Fig. 4(f), lacking prior knowledge, starts from a low FoM and struggles to improve efficiently.

4.5 Case Study

To further analyze the internal mechanism of ASL-BO, we present a case study on the sizing task of a two-stage Miller compensation op-amp. In topological equivalence analysis, the LLM agent parses the netlist to identify the differential input pair, active load current mirror and tail current source, and imposes six equality constraints based on these symmetric pairs: $W_{\text{diff}} = W_{\text{diff}1}$, $L_{\text{diff}} = L_{\text{diff}1}$, $W_{\text{load}} = W_{\text{load}1}$, $L_{\text{load}} = L_{\text{load}1}$, $W_{\text{tail}} = W_{\text{tail}1}$, $L_{\text{tail}} = L_{\text{tail}1}$. It reduces design variables from 16 transistor parameters to 10 independent variables plus compensation components, effectively pruning the search space.

In heuristic initialization, the agent generates 10 initial samples covering baseline balance, high-gain long-channel, zero-resistance and boundary stress tests according to design specifications and process constraints. Over 10 iterations, it refines boundaries adaptively: narrowing width bounds and fixing the channel lengths to $1 \mu\text{m}$ for higher gain in iterations 1–4 with the qKG acquisition function, then tuning the remaining parameters to the optimal point and switching to qLogEI for confirmation in subsequent iterations.

5 Conclusion

We effectively integrated the complementary strengths of LLM-based agent and Bayesian optimization (BO), and applied them to analog circuit sizing. Specifically, we put forward a novel method ASL-BO. Experimental results on two typical analog circuits across four specifications proved that ASL-BO achieves both high accuracy and efficiency in analog circuit sizing.

The limitations of the current work lie in circuit scale and model diversity:

1. The current ASL-BO is only validated on operational amplifiers and hysteresis comparators, and has not been tested on large-scale, hierarchical analog mixed-signal systems. It cannot fully demonstrate its optimization effect and stability in complex circuits like ADCs, PLLs, and other system-level circuits.
2. The experimental verification of ASL-BO is only based on DeepSeek-V3.2 and GPT-4.1, without accessing more open-source small-parameter models or domain-specific fine-tuned models. The adaptability and performance differences of the method under different models have not been systematically explored.

Future work can refine ASL-BO on larger-scale hierarchical systems like ADCs, and explore knowledge distillation, finetuning or targetedly designed BO acquisition strategy to further reduce computational cost.

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